INCH-POUND

MIL-PRF-32517
24 March 2015

PERFORMANCE SPECIFICATION

SPLICE, CONDUCTOR, ELECTRICAL, COAXIAL CABLE, MATCHED IMPEDANCE, GENERAL SPECIFICATION FOR

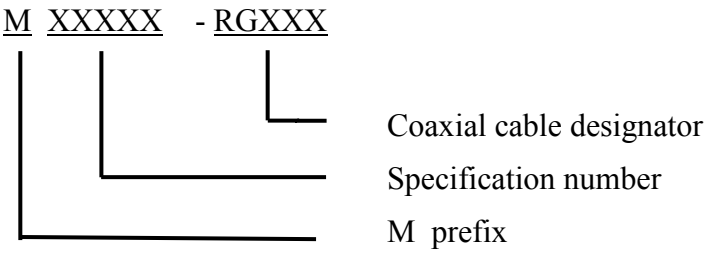
This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the minimum requirements for an environment resistant, permanent splice with a temperature rating of -55 °C to 150 °C with an environmental seal and matched impedance to repair coaxial cables.

1.2 Part or Identifying Number (PIN). The PINs to be used for matched impedance splices acquired to this specification are created as follows. Part number designations are listed in Table I.

Example:



Comments, suggestions, or questions on this document should be addressed to the Naval Air Systems Command, (Commander, Naval Air Warfare Center, Aircraft Division, Code 412000B120-3, Highway 547, Joint Base MDL, NJ 08733-5100) or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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TABLE I. Part number designations.

Splice Part Number	Cable	MIL-DTL-17 Specification Sheet
M32517-RG142	RG142	/60
M32517-RG179	RG179	/94
M32517-RG302	RG302	/110
M32517-RG316	RG316	/113
M32517-RG393	RG393	/127
M32517-RG400	RG400	/128

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

P-D-410 - Dishwashing Compound, Hand, (Synthetic Detergent, Solid And Liquid Form)

FEDERAL STANDARD

FED-STD-228 - Test Methods for Cables and Wire, Insulated

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DEPARTMENT OF DEFENSE STANDARD

MIL-STD-202	-	Electronic And Electrical Component Parts

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-17	-	Cables, Radio Frequency, Flexible and Semirigid, General Specification for
MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 And JP-5
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-152, O-154, O-156 and O-167
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-PRF-87937	-	Cleaning Compound, Aerospace Equipment

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other government documents. The following other government document forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of this document is that cited in the solicitation or contract.

STANDARDIZATION DOCUMENT

SD-6	-	Provisions Governing Qualification

(Copies of this document are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA-364-101	-	Attenuation Test Procedure for Electrical Connectors, Sockets, Cable Assemblies or Interconnection Systems
ECIA EIA-364-108	-	Impedance, Reflection Coefficient, Return Loss, and VSWR Measured in the Time and Frequency Domain Test Procedure for Electrical Connectors, Cable Assemblies or Interconnect Systems

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(Copies of these documents are available online at http://www.ecianow.org.)

SAE INTERNATIONAL

SAE AMS1424	-	Fluid, Deicing/Anti-icing, Aircraft SAE Type I
SAE AS22520	-	Crimping Tools, Wire Termination, General Specification for
SAE AMS-DTL-23053	-	Insulation Sleeving, Electrical, Heat Shrinkable, General Specification for

(Copies of these documents are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Matched impedance splices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Cable. Matched impedance splices shall be compatible with the cables conforming to the electrical performance requirements of MIL-DTL-17.

3.3 Matched impedance splice assembly performance. The matched impedance splice assembly shall conform to the requirements specified herein.

3.3.1 Sealing sleeve. If a sealing sleeve is part of the splice assembly, the sealing sleeve component of the splice assembly shall conform to the longitudinal change, uninstalled recovery, and copper corrosion requirements of SAE AMS-DTL-23053.

3.3.2 Metal crimp splice. When a metal crimp barrel is used to join the inner conductor or shielding of the MIL-DTL-17 cable, SAE AS22520 crimp tools, dies and gages shall be used to install the metal crimp barrel. If an existing SAE AS22520 crimp tool is not available, the manufacturer shall request authorization from the procuring activity to provide a non SAE AS22520 tool.

3.3.3 Design. The matched impedance splice shall meet the maximum dimensions of Table II and Figure 1.

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TABLE II. Maximum dimensions.

Matched impedance splice part number	D (inches) Maximum	L (inches) Maximum
M32517-RG142	.393 (10 mm)	2.56 (65 mm)
M32517-RG179	---	---
M32517-RG302	---	---
M32517-RG316	---	---
M32517-RG393	.630 (16mm)	2.56 (65 mm)
M32517-RG400	.433 (11 mm)	2.56 (65 mm)

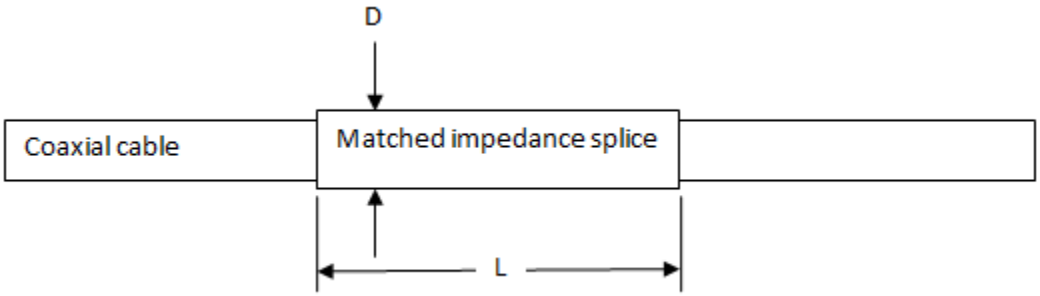


FIGURE 1. Matched impedance splice dimensions.

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3.4 Performance characteristics.

3.4.1 Characteristic impedance. When matched impedance splice assemblies are tested as specified in 4.6.1, the characteristic impedance shall not exceed the impedance limits shown in Table III.

TABLE III. Characteristic impedance.

Matched impedance splice part number	Impedance (Ohms)
M32517-RG142	48 - 55
M32517-RG179	
M32517-RG302	
M32517-RG316	
M32517-RG393	46 – 57.5
M32517-RG400	48 - 55

3.4.2 Insertion loss. When matched impedance splice assemblies are tested as specified in 4.6.2, the insertion loss shall not exceed one decibel (dB) above the limits throughout the frequency range up to 3 gigahertz (GHz) as specified in applicable cable MIL-DTL-17 specification sheet (see 6.5). At the limit of the frequency range for MIL-DTL-17 cables operating above 3 GHz, the insertion loss limit shall increase linearly to 3dB above the MIL-DTL-17 limits, at the highest rated frequency (see Table IV).

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TABLE IV. Matched impedance splice performance. 1/

Frequency (MHz)	Insertion loss 10 feet (dB)	Return loss (dB)	Standing Wave Ratio (SWR) 2/
M32517-RG142 Matched impedance splice (MIL-DTL-17/60 Cable)			
50	1.6	26	1.099
100	1.6	25	1.122
400	1.8	24	1.133
1,000	2.3	23	1.152
2,000	3.0	21	1.196
3,000	3.7	21	1.196
8,000	8.1	21	1.196
12,400	11.4	21	1.196
M32517-RG179 Matched impedance splice ((MIL-C-17/94 Cable)			
400	3.1	Not Applicable	Not Applicable
M32517-RG302 Matched impedance splice (MIL-C-17/110 Cable)			
400	1.8	Not Applicable	Not Applicable
3,000	3.6		
M32517-RG316 Matched impedance splice (MIL-C-17/113 Cable)			
50	1.75	28	1.083
100	2.1	28	1.083
400	3.1	21	1.196
1,000	6.8	19	1.253
3,000	---	15	1.433
M32517-RG393 Matched impedance splice (MIL-C-17/127 Cable)			
50	1.17	23	1.152
100	---	23	1.152
400	1.5	---	---
1,000	---	20	1.196
3,000	2.8	18	1.288
5,000	---	11	1.433
11,000	7.5	10	1.498

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TABLE IV. Matched impedance splice performance – Continued. 1/

Frequency (MHz)	Insertion Loss 10 feet (dB)	Return Loss (dB)	SWR 2/
M32517-RG400 Matched impedance splice (MIL-DTL-17/128 Cable)			
50	1.32	23.5	1.148
100	1.45	22.0	1.172
400	2.05	21.8	1.177
1,000	2.70	20.0	1.222
2,000	4.00	18.0	1.288
3,000	4.80	17.0	1.329
10,000	10.40	16.0	1.329
12,400	12.00	16.0	1.329

1/ Matched impedance splice performance was determined using MIL-DTL-17 cables with cable leads at least 5 feet in length (10 feet total length).

2/ SWR is mathematically calculated from the return loss (see 6.4.3).

3.4.3 Return loss. When cables are tested as specified in 4.6.3, the return loss shall not exceed the limits of table IV. SWR shall be determined by cross-referencing the adjusted return loss value to SWR using the applicable table in the detail specification.

3.4.4 Voltage drop. When installed the matched impedance splice assemblies are tested as specified in 4.6.4, the voltage drop across the shield termination shall not exceed that of an equivalent length of the cable by more than 1.5 millivolts (before conditioning) and 2.0 millivolts (after conditioning). When installed inner conductor splices are tested in accordance with 4.6.4, the voltage drop shall not exceed that of an equivalent length of wire by more than 2.0 millivolts (before conditioning) and 2.5 millivolts (after conditioning).

3.4.5 Insulation resistance. When installed the matched impedance splice assemblies are tested in accordance with 4.6.7, the insulation resistance shall be no less than 5000 mega-ohms.

3.4.6 Dielectric withstanding voltage. When matched impedance splice assemblies are tested as specified in 4.6.8, there shall be no breakdown, flashover, or spark-over. The matched impedance splice shall meet the dielectric withstanding voltage requirements in applicable cable MIL-DTL-17 detail specification sheet. The leakage current shall be less than 2.5 milliamperes maximum.

3.4.7 Tensile strength. The inner conductor shall not separate from the matched impedance splice when the installed matched impedance splice is tested in accordance with 4.6.5. The minimum tensile strength is specified in Table V.

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TABLE V. Tensile strength requirement.

Matched impedance splice part number	Tensile strength (Pounds minimum)
M32517-RG142	85
M32517-RG179	3
M32517-RG302	40
M32517-RG316	9
M32517-RG393	125
M32517-RG400	8

3.4.8 Thermal shock. When installed matched impedance splice assemblies are subjected to five thermal shock cycles between -55 °C and +150 ºC in accordance with 4.6.6.3, there shall be no evidence of damage to the termination, and shall meet the voltage drop and tensile strength requirements specified in 4.6.4 and 4.6.5. Discoloration of the device insulation shall not be cause for rejection.

3.4.9 Altitude immersion. When installed matched impedance splice assemblies are subjected to three altitude immersion cycles between sea level and 75,000-foot altitude in accordance with 4.6.6.1, splice assemblies shall meet the insulation resistance requirements as described in 4.6.7.

3.4.10 Fluid immersion. When installed splice assemblies are exposed to any of the six test fluids in accordance with 4.6.6.2, the terminations shall meet the dielectric withstanding voltage (see 4.6.8) requirements in the applicable cable MIL-DTL-17 specification sheet.

3.4.11 Moisture resistance. The splice assembly shall be tested in accordance with 4.6.6.4. There shall be no evidence of water intrusion in the matched impedance splice.

3.4.12 Vibration. The splice assembly shall be subjected to vibration in accordance with 4.6.9. There shall be no evidence of cracking, breaking or loosening of the splice assembly and the termination shall meet the voltage drop and tensile strength requirements specified in 4.6.4 and 4.6.5.

3.4.13 Heat aging. When installed, the matched impedance splice assembly is subjected to heat aging at 165 °C in accordance with 4.6.10. The matched impedance splice assembly shall meet the subsequent performance requirements in table IV. Discoloration of the device insulation shall not be cause for rejection.

3.4.14 Flammability. The matched impedance splice shall be self-extinguishing within 5 seconds after the removal from the flame in accordance with 4.6.11. Self-extinguishing above 5 seconds due to the MIL-DTL-17 cable is not considered to be a failure.

3.4.15 Copper mirror corrosion. When matched impedance splice assemblies are tested for 16 hours at 121 °C in accordance with 4.6.12, copper removal shall not exceed 10 percent of the area of the mirror above the bottom 0.063 inch.

3.5 Workmanship. The splice assembly components shall be uniform in quality and shall be free of defects detrimental to life, serviceability, or performance.

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3.6 Heat source. The manufacturer's recommended heat convection air, infrared, or equivalent shall be used.

3.7 Assembly instructions. If specified in the contract (see 6.2), assembly instructions for each matched impedance splice assembly kit will include:

a. Cable preparation – stripping dimensions and tolerances.
b. List and description of crimping, stripping, or special tools if required.
c. Pictorial presentation of subassemblies and loose piece parts.
d. Sufficient pertinent dimensions for verification of correct parts: As a minimum, the cable entry openings for the conductor, dielectric, braid, and jacket should be specified.
e. Military PIN and manufacturer's PIN (see 1.2).

4. VERIFICATION

4.1 Classification of inspections. The examinations and tests of matched impedance splice assemblies are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made at ambient temperature and humidity as specified in MIL-STD-202, 4.2 Test conditions.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.2.2 Matched impedance splice assembly. The splices shall be installed on cables conforming to MIL-DTL-17 by the testing activity using the specified tooling (see 3.3.2). Unless otherwise specified, the leads shall be at least 5 feet in length (10 feet total length).

4.2.3 Temperature stabilization. Voltage drop measurements shall be made after the temperature of the matched impedance splice assembly has stabilized. Temperature stabilization shall be determined by 3 consecutive readings within ±1 °C at intervals of 3 minutes each. All tests performed after exposure to high or low temperature shall be conducted after splices have been conditioned for at least 1 hour at the inspection conditions specified (see 4.2).

4.2.4 Water bath. Unless otherwise specified in the applicable test method, a water bath containing 0.5 percent of an anionic wetting agent (see P-D-410) and 5.0 percent sodium chloride shall be used whenever immersion is specified. Free ends of coaxial cable shall be a minimum of 4 inches from top surface of water.

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests in table VI. Unless otherwise specified in table VI, sequential testing is required for each group. The qualifying activity shall perform or witness the particular tests as specified in table VI and the supplier shall perform the remaining tests.

4.3.1 Qualification inspection process. At the initiation of the qualification process, prior to testing, prospective suppliers must forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process is described in the SD-6 (see 6.3.1).

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4.3.2 Qualification inspection samples. Matched impedance splices shall be assembled as specified in 4.2.2 for testing and submitted to the qualifying activity as follows:

a. For the qualifying activity tests, 25 unwired specimens (19 for testing and 6 spares) for each part number shall be separately packaged and forwarded to the address (see 6.3) designated in the letter of authorization. Sufficient cables conforming to MIL-DTL-17 shall be provided to perform the designated tests. Each sample shall be identified by marking each package with the following information: Sample for qualification

Specification MIL-PRF-32517
Part number _____
SPLICE, CONDUCTOR, ELECTRICAL, COAXIAL CABLE, MATCHED IMPEDANCE
Name of supplier
Supplier's part number
Submitted (date) under authorization (reference letter authorizing the inspection)

b. For the supplier tests, 25 specimens shall be tested in accordance with table VI for each part number.

4.3.3 Retention of qualification. Retention of qualification inspections shall be performed every 36 months after the original qualification approval date. The qualifying activity shall perform or witness the tests specified in table VII. Unless otherwise specified in table VII, sequential testing is required for each group. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

4.3.3.1 No production certification. If no production of the qualified products has occurred for the entire reporting period, the supplier shall provide certification to the qualifying activity that no changes in the product materials, design, manufacturing processes, or site of production have occurred since qualification approval (see 4.3). Certification for more than one reporting period is not permitted. The supplier shall contact the qualifying activity to determine the conditions for certification. The supplier shall submit the certification on DD Form 1718 provided by the qualifying activity.

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TABLE VI. Qualification inspections.

Examination or test	Requirement paragraph	Test method paragraph
Group I (19 specimens) 1/, 2/, 3/		
Visual examination	3.3.1, 3.3.2	4.5.1
Workmanship	3.5	4.5.1
Group II (1 uninstalled sealing sleeve)		
Copper mirror corrosion	3.4.15	4.6.12
Group III (3 matched impedance splice assemblies) 1/, 3/		
Characteristic impedance	3.4.1	4.6.1
Insertion loss	3.4.2	4.6.2
Return loss	3.4.3	4.6.3
Dielectric withstanding voltage (dry)	3.4.6	4.6.8.1
Group IV (3 matched impedance splice assemblies)		
Voltage drop (initial)	3.4.4	4.6.4
Thermal shock	3.4.8	4.6.6.3
Voltage drop (after test)	3.4.4	4.6.4
Group V (4 matched impedance splice assemblies) 1/, 2/		
Environmental conditioning		
Altitude immersion (3 cycles)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Immersion	---	4.6.6.5
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Thermal shock	3.4.8	4.6.6.3
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Moisture resistance	3.4.11	4.6.6.4
Vibration	3.4.12	4.6.9
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Heat aging	3.4.13	4.6.10
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Insertion loss	3.4.2	4.6.2
Return loss	3.4.3	4.6.3
Dielectric withstanding voltage (wet)	3.4.6	4.6.8.2
Voltage drop (after test)	3.4.4	4.6.4
Tensile strength	3.4.7	4.6.5

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TABLE VI. Qualification inspections – Continued.

Examination or test	Requirement paragraph	Test method paragraph
Group VI (2 matched impedance splice assemblies) 1/		
Flammability	3.4.14	4.6.11
Group VII (6 matched impedance splice assemblies) 1/		
Fluid immersion	3.4.10	4.6.6.2
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Dielectric withstanding voltage (Wet)	3.4.6	4.6.8.2

1/ The matched impedance splice assemblies shall be installed on MIL-DTL-17 cables with cable leads at least 5 feet in length (10 feet total length). See 4.2.2.
2/ Group sequence not required (two separate specimens).
3/ Qualifying activity test groups

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TABLE VII. Retention of qualification inspections.

Examination or Test	Requirement paragraph	Test method paragraph
Group I (10 specimens)		
Visual examination	3.3.1, 3.3.2	4.5.1
Workmanship	3.5	4.5.1
Group III (3 matched impedance splice assemblies) 1/		
Characteristic impedance	3.4.1	4.6.1
Insertion loss	3.4.2	4.6.2
Return loss	3.4.3	4.6.3
Dielectric withstanding voltage (dry)	3.4.6	4.6.8.1
Group IV (3 matched impedance splice assemblies)		
Voltage drop (initial)	3.4.4	4.6.4
Thermal shock	3.4.8	4.6.6.3
Voltage drop (after test)	3.4.4	4.6.4
Group V (4 matched impedance splice assemblies) 1/		
Environmental conditioning		
Voltage drop (pre-test)	3.4.4	4.6.4
Altitude immersion (3 cycles)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Immersion	---	4.6.6.5
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Thermal shock	3.4.8	4.6.6.3
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Moisture resistance	3.4.11	4.6.6.4
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Heat aging	3.4.13	4.6.10
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Insertion loss	3.4.2	4.6.2
Return loss	3.4.3	4.6.3
Dielectric withstanding voltage (wet)	3.4.6	4.6.8.2
Voltage drop (after test)	3.4.4	4.6.4
Tensile strength	3.4.7	4.6.5

1/ The matched impedance splice assemblies shall be installed on MIL-DTL-17 cables with cable leads at least 5 feet in length (10 feet total length). See 4.2.2.

4.4 Conformance inspection. Conformance inspection shall consist of Group A examinations and test (see table VIII) and Group B inspections (see table X).

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4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of Group A inspections. Except as specified in 4.4.3.2, delivery of products that have passed Group A inspection shall not be delayed pending the results of Group B inspection.

4.4.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of matched impedance splices manufactured under essentially the same conditions and offered for inspection at one time.

4.4.2 Group A inspection. Group A inspection shall consist of the examination and test specified in table VIII and shall be made on the same set of sample units, in the order shown.

4.4.2.1 Sampling plan. Statistical sampling shall be in accordance with table IX. A randomly selected group(s) of samples is required. For acceptance of the lot there shall be zero occurrences of defects.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the supplier shall withdraw the lot and may rework it to correct the defects or screen out the defective units. Such lots shall be kept separate from new lots and shall be re-inspected. Such lots shall be re-inspected using tightened inspection.

TABLE VIII. Group A inspections.

Examination or test	Requirement paragraph	Test method paragraph	Sampling Plan
Visual examination	3.3.1, 3.3.2	4.5.1	4.4.2.1
Workmanship	3.5	4.5.1	
Tensile strength	3.4.7	4.6.5	

1/ At the option of the manufacturer, documented in-process inspection may be used to satisfy the requirements provided that all of the manufacturer's in-process control data on these tests are made available to the Government upon request.

TABLE IX. Zero defect sampling plan.

Lot size	Minimum number of matched impedance splices to be tested 1/
1 – 4	All
5 – 50	5
51 – 90	7
91 – 150	11
151 – 280	13
281 – 500	16
501 – 1200	19
1201 – 3200	23
3201 – 10000	29
10001 – 35000	35

1/ For lot sizes that are less than the required sample size, all matched impedance splices shall be tested.

4.4.3 Group B inspection. Group B inspection shall consist of the tests specified in table X. Group B inspection shall be made on matched impedance splices selected from inspection lots that have passed Group A inspections.

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TABLE X. Group B inspections.1/

Examination or test	Requirement paragraph	Test method paragraph
Characteristic impedance	3.4.1	4.6.1
Insertion loss	3.4.2	4.6.2
Return loss	3.4.3	4.6.3
Altitude immersion (1 cycle)	3.4.9	4.6.6.1
Insulation resistance	3.4.5	4.6.7
Dielectric withstanding voltage (wet)	3.4.6	4.6.8.2
Voltage drop	3.4.4	4.6.4

1/ A manufacturer's normal control tests and production tests may be used to fulfill all or part of Group B inspection; however, all of the Group B inspections shall be completed as specified.

4.4.3.1 Sampling plan. Six months after the date of notification of qualification, and after each subsequent 6 month period, fifteen splices shall be selected.

4.4.3.2 Failures. If one or more matched impedance splices fail to pass Group B inspection, the sample shall be considered to have failed. The supplier shall take corrective action on the materials or process, or both, as warranted and on all splices that are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action has been taken. After the corrective action has been taken, Group B and/or Group A (see table VIII) inspection, as applicable, shall be repeated on additional splices. Final acceptance shall be withheld until Group B re-inspection, as applicable, has shown that the corrective action was successful. The corrective action shall be submitted to the qualifying activity as part of the retention of qualification (see 4.3.3).

4.4.3.3 Disposition of samples. Samples that have been subjected to Group B inspection shall not be delivered on the contract or order.

4.5 Examinations.

4.5.1 Visual and dimensional examination. Matched impedance splices shall be examined to verify that the materials, design, construction, and physical dimensions are in accordance with this specification (see table VI). Classification of defects for visual and dimensional examination shall be as specified in table XI. No major or minor or defects are permitted for qualification testing.

a. Dimensions shall be measured with calibrated micrometers using standard laboratory methodology.

b. Sleeve thickness (when specified): The sleeving shall be split lengthwise into two hemi cylindrical pieces. A smooth gage rod shall be selected, measured and placed in contact with the inside surface of the sleeving. The total thickness of the gage rod plus sleeving wall shall then be measured. The maximum and minimum wall thickness shall be calculated by subtracting the gage rod measurement from the measurement of the gage rod plus sleeving wall.

c. Identification of product: Visually examine for color coding.

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d. Workmanship: Visually examine with a 3X microscope for blistering, pitting, or peeling of plating, cracks, and other defects which may affect serviceability.

TABLE XI. Classification of defects.

Examination or test	Major	Minor
Workmanship and markings		X
Missing components	X	
All other criteria		X

4.6 Test methods.

4.6.1 Characteristic impedance (see 3.4.1).

4.6.1.1 Specimen. The specimen shall be not less than 10 feet.

4.6.1.2 Procedure. The specimen shall be prepared for testing by assembling appropriate connections to the cable ends. The equipment shall include a Time Domain Reflectometer (TDR). The rise time of the TDR shall be 150 picoseconds or less, and the vertical sensitivity of the system shall provide for a minimum resolution of one major scale division. Matched impedance splice measurement shall be taken by gating within 1 nanosecond on each side of the splice.

The matched impedance splice assembly shall then be turned end-to-end and the measurement repeated. For cables of other than 50 and 75 ohms characteristic impedance, the characteristic impedance may be determined by calculation from the capacitance measurement determined and the velocity of propagation measurement determined, using the following formula:

Z_o = Velocity of propagation (%) X Capacitance (pF/ft)

4.6.2 Insertion loss. The insertion loss shall be determined in accordance with EIA-364-101, Method A. A vector network analyzer is preferred. For cable impedances other than 50 ohms, minimum loss impedance matching pads or a mechanical adapter with adapter characteristic embedding may be used with open/short/load calibration standards for the cable impedance value (see 3.4.2 and 6.5).

4.6.3 Return loss. The return loss shall be measured in accordance with EIA-364-108, frequency domain method. For cable impedances other than 50 ohms, minimum loss impedance matching pads or a mechanical adapter with adapter characteristic embedding may be used with open/short/load calibration standards for the cable impedance value. SWR is mathematically calculated from the return loss (see 3.4.3).

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4.6.4 Voltage drop. Splices shall be tested as follows (see 3.4.4):

a. Sample preparation. The MIL-DTL-17 cable insulation shall be stripped 2 inches from both ends. The shield shall be trimmed so exposed length is ¼ inch. Strip ¼ inch of the primary conductor's insulation.

b. Outer conductor. Measurements shall be made by puncturing the insulation of the current carrying conductor on each end of the splice ¼ inch back from the ends of the matched impedance splice. The distance between the 2 test points shall be noted. Measurement of the current carrying conductor shall be made by puncturing the conductor insulation the same distance between test points as that noted for the splice measurement. The millivolt drop of the equivalent length of wire may be determined by averaging 4 readings taken on 10-inch lengths of wire selected at random throughout the supply of wire to be used for subsequent tests. The millivolt drop through the crimp termination and the current carrying conductor shall be measured while 1 ampere direct current (DC) is being applied.

c. Inner conductor. Measurements shall be made by puncturing the insulation of the current carrying conductor on each end of the splice 3½ inches back from the ends of the matched impedance splice. The distance between the 2 test points shall be noted. Measurement of the current carrying conductor shall be made by puncturing the conductor insulation the same distance between test points as that noted for the splice measurement. The millivolt drop of the equivalent length of wire may be determined by averaging 4 readings taken on 10 inch lengths of wire selected at random throughout the supply of wire to be used for subsequent tests.

4.6.5 Tensile strength. Stripped ends of the repaired cable shall be placed in a standard tensile-testing machine so that the matched impedance splice is centered between and at least 3 inches from the jaws. Sufficient force shall be applied to pull the inner conductor out of the splice or break the cable or the splice. The travel speed of the head shall be 1 inch per minute. The clamping surfaces of the jaws may be serrated to provide sufficient force (see 3.4.7).

4.6.6 Environmental conditioning. The matched impedance splices shall be exposed to each of the conditions in the sequence shown in table VI.

4.6.6.1 Altitude immersion. The installed matched impedance splice shall be immersed in a water bath and placed in a suitable chamber; the free ends shall be within the chamber and shall not be sealed (see 4.2.4). The chamber pressure shall be reduced to 75,000 ft. (26.78 mm of Hg) and maintained for 30 minutes. The chamber shall then be returned to ambient pressure (see 4.2.). This shall constitute 1 cycle. A total of 3 cycles shall be run. The insulation resistance shall be measured after the third cycle (see 3.4.9).

4.6.6.2 Fluid immersion. The installed matched impedance splice assemblies shall be immersed in the fluids specified in table XII at the temperature and time period listed. A separate matched impedance splice assembly shall be immersed in each of the required fluids. All fluid immersion samples shall be subjected to the subsequent tests (see 3.4.10).

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TABLE XII. Fluid immersion.

Specimen number	Test fluid specifications	Test conditions
1 2	MIL-PRF-7808 MIL-PRF-23699	(a) Immerse splice in fluid at 120 °C ±3 °C for 5 minutes. (b) Remove splice and allow to drain for 1 hour at 23 °C ±5 °C (c) Place splice in a 175 °C ±3 °C oven for 22 hours. (d) Repeat (a), (b), and (c) for 7 cycles.
3	MIL-PRF-5606	Same as above, except immerse in fluid at 85 °C ± 3 °C for step (a). Expose sample to 104 °C ± 3 °C for step (c).
4 5	SAE AMS1424 MIL-PRF-87937 (Undiluted)	Same as MIL-PRF-7808 except immerse in fluid at 65 °C ± 3 °C for step (a).
6	MIL-DTL-5624 JP-5 or MIL-DTL-83133 JP-8	Immerse splice in fluid at 23 °C ±5 °C for 20 hours. Remove and drain for 4 hours at 23 °C ±5 °C.

4.6.6.3 Thermal shock. Installed matched impedance splices shall be conditioned in accordance with MIL-STD-202, Method 107, test condition F, except that the maximum temperature extreme shall be the maximum continuous operating temperature of 150 °C or the maximum operating temperature of the MIL-DTL-17 cable, whichever is lower (see 3.4.8).

4.6.6.4 Moisture resistance. The matched impedance splices shall be tested in accordance with MIL-STD-202, Method 106, except sub cycle 7b shall not be required (see 3.4.11).

4.6.6.5 Immersion. The matched impedance splices shall be tested in accordance with MIL-STD-202, Method 104, test condition C.

4.6.7 Insulation resistance. Installed matched impedance splices shall be immersed in a water bath as specified in 4.2.4 for at least 30 minutes. Matched impedance splices shall be tested in accordance with MIL-STD-202, Method 302. The following details shall apply (see 3.4.5):

a. Test condition: A
b. Electrification time: 1 minute
c. Points of measurement: Between cable shield and water bath, between each conductor and cable shield.

4.6.8 Dielectric withstanding voltage (see 3.4.6).

4.6.8.1 Dry dielectric. The installed matched impedance splice shall be tested in accordance with FED-STD-228, Method 6111, except that the specimen shall not be immersed in water but tested dry. The specified dielectric withstanding voltage (see 3.4.6) at a frequency of 60 hertz (Hz) shall be applied for a minimum of 1 minute. The test voltage shall be applied between the inner and the outer conductor, with the outer conductor grounded.

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4.6.8.2 Wet dielectric. The installed matched impedance splice shall be immersed in a water bath as specified in 4.2.4 for at least 30 minutes. The installed matched impedance splice shall be tested in accordance with MIL-STD-202, Method 301. The following details shall apply:

a. Dielectric withstanding voltage: See 3.4.6.
b. Electrification time: 1 minute.
c. Points of measurement: Between cable shield and water bath, between each conductor and cable shield.

4.6.9 Vibration. A complete matched impedance splice assembly shall be mounted as shown in figure 2 and vibrated in accordance with of MIL-STD-202, test method 204 (see 3.4.12). One lead of the matched impedance splice shall be rigidly mounted by a suitable clamp to a test fixture securely fastened to the vibration platform. The opposite lead of the matched impedance splice assembly shall be clamped to a stationary support so that the center of the matched impedance splice is 6 inches from the vibration and stationary platforms, and all slack is removed from the matched impedance splice leads with minimum tension on the matched impedance splice assembly. The shield and center conductor shall be connected to a suitable monitoring device. At least 100 milliamperes shall be flowing through matched impedance splice assembly. The following conditions shall apply:

a. Test condition letter B.
b. Continuity shall be monitored during vibration with a detector capable of detecting interruptions of 1 microsecond duration or longer, or as specified at 100 milliamperes.

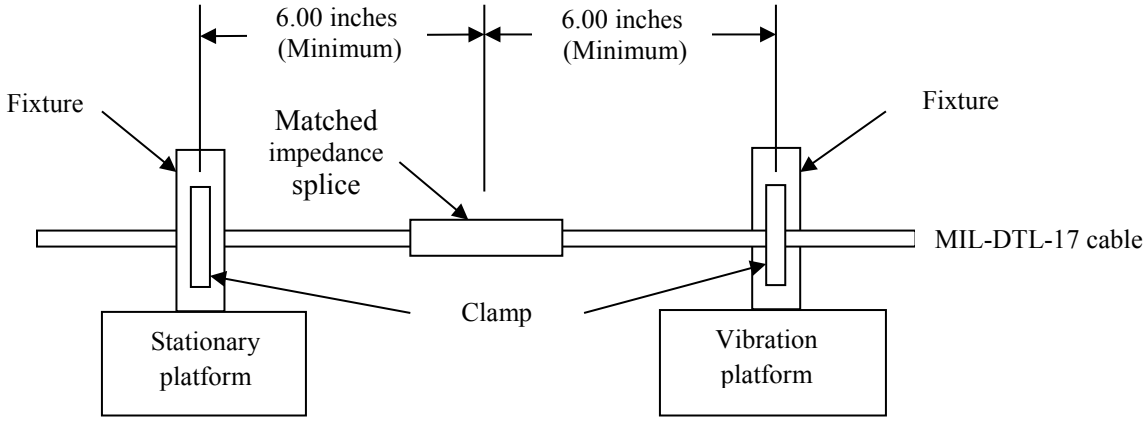


FIGURE 2. Vibration testing setup.

4.6.10 Heat aging. The installed matched impedance splice shall be placed in a forced-convection oven, having a circulating air velocity of 100 to 200 feet per minute, so that the major axis of the sleeve is vertical and no sleeve is in contact with any of the other sleeves (see 3.4.13). The samples shall be conditioned at a temperature of 165 °C or the maximum operating temperature of the MIL-DTL-17 cable, whichever is lower, for a period of 750 hours. After removal from the chamber, the splices shall be stabilized (see 4.2.3).

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4.6.11 Flammability. The splices shall be suspended horizontally in a draft-free enclosure above a Bunsen burner. The tip of a 1 inch natural gas flame with an inner core 1/3 its height shall be applied to the end of the sealing sleeve for a period of 20 seconds. The Bunsen burner shall have a ¼-inch inlet, a nominal bore of 3/8 inch, and a length of approximately 4 inches above the primary inlets. Self-extinguishing above 5 seconds due to the MIL-DTL-17 cable is not considered to be a failure (see 3.4.14).

4.6.12 Copper mirror corrosion. For sizes 0.300 inch and larger (as supplied I.D.) the termination shall be cut longitudinally into 4 sections. For sizes smaller than 0.300, the whole terminations shall be used. The termination or all pieces of cut terminations shall be placed in a clean, dry ½ by 12 inch test tube. A copper glass mirror about ¼ by 1 inch and a calibrated thermometer shall be suspended 6 to 7 inches above the bottom of the test tube by means of a fine, clean copper wire attached to a cork that has been wrapped in clean aluminum foil. A second clean test tube containing only the mirror and thermometer shall be used as a control. The lower 2 inches of the test tube shall be immersed in an oil bath at 121 ±3 °C for 16 ±0.5 hours. The temperature of that portion of the test tube containing the copper mirror shall be below 60 °C. After removal from the oil bath and cooling to ambient temperature (see 4.2), visually examine the mirror for areas of transparency by placing it on a white background in good light. The mirrors shall be vacuum deposited copper with a thickness equal to 10 ±5 percent transmission of normal light of 5000 Angstroms. They shall be stored in a vacuum and shall be used only if no oxide film is present and the copper is not visibly damaged or contaminated (see 3.4.15).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. Matched impedance splices covered by this specification are for use in making environmentally protected permanent joints on MIL-DTL-17 coaxial cables. The matched impedance splice is military unique because it is required to match the impedance of MIL-DTL-17 coaxial cables to reduce insertion loss and maintain signal integrity. In addition, the matched impedance splice must work in the military aircraft operating environment, which requires environment resistance and the ability to operate in a high electromagnetic interference, high shock, high vibration, high and low temperature and high corrosion environment. A commercial substitute is not available.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. PIN (see 1.2).
c. Assembly instructions, if required (see 3.7).
d. Packaging (see 5.1).
e. Special markings, if required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List (QPL) No. 32517 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the qualifying activity, Naval Air Systems Command (Code 4.4.5.3), 48298 Shaw Road, Building 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Provisions governing qualification. Qualification will be conducted in accordance with the provisions and guidelines of SD-6, "Provisions Governing Qualification." Copies of this document are available online at https://assist.dla.mil.

6.3.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible website http://www.navair.navy.mil/qpl/.The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested and the manufacturing location of the component.

6.4 Definitions.

6.4.1 Critical defects. A critical defect is a defect that judgment and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgment and experience indicate is likely to prevent performance of the actual function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

6.4.2 Defect. A defect is any nonconformance of the splice with specified requirements.

6.4.2.1 Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce the usability of the splice for its intended purpose.

6.4.2.2 Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the splice for its intended purpose, or is a departure from established standards having little bearing on the effective use of the splice.

6.4.3 Standing Wave Ratio. Standing Wave Ratio (SWR) is a mathematical expression of the non-uniformity of an electromagnetic field (EM field) on a transmission line such as

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coaxial cable. Usually, SWR is defined as the ratio of the maximum radio-frequency (RF) voltage to the minimum RF voltage along the line. SWR may be calculated from the return loss (RL) using the following equation.

$$SWR = (1+(10^{RL/20}))/((10^{RL/20})-1))$$

6.5 Insertion loss example. The following figure is provided as an example to show the matched impedance splice attenuation relative to the MIL-DTL-17 cable attenuation. In this example, the performance is based on matched impedance splice leads using MIL-DTL-17 cables that are 5 feet (total length both leads 10 feet).

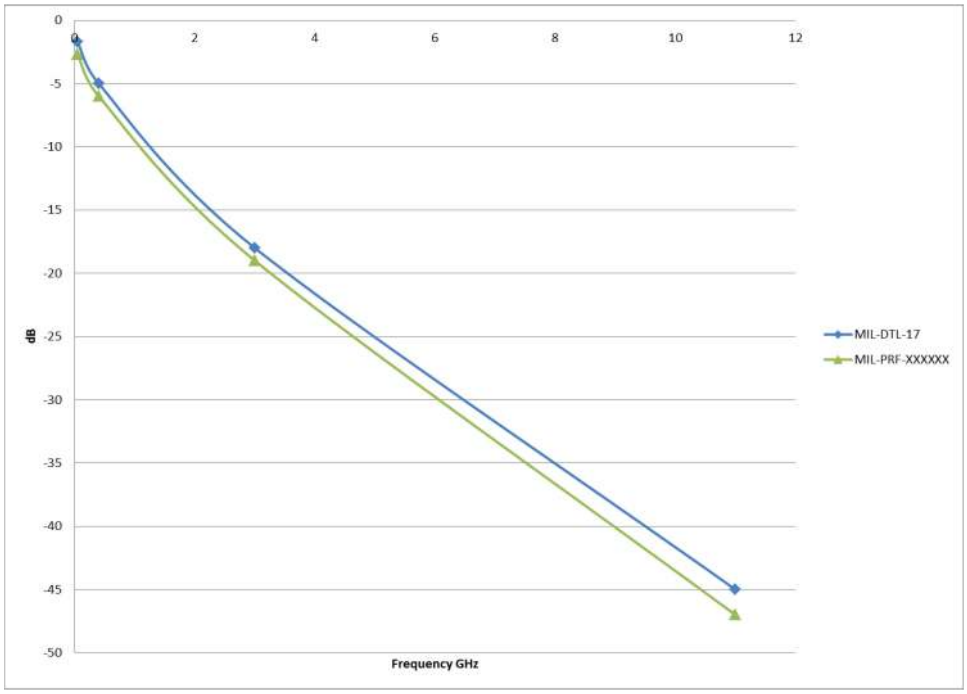


FIGURE 3. Insertion loss limit(s) for RG393 (MIL-C-17/127).

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6.6 Subject term (key word) listing.
Crimp
Copper mirror corrosion
Flammability
Heat aging
Insertion loss
Radio frequency (RF)
Sealing sleeve
Standing wave ratio

Custodians:
Navy – AS
Air Force – 85

Preparing activity:
Navy – AS
Project 5940-2015-001

Review activity:
Air Force - 99

NOTE: The activity listed above was interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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